

- 7 Female mosquitoes feed on the blood of other animals to promote the development of their eggs. During this process, infected mosquitoes can transmit the *Plasmodium* parasite.

The *Plasmodium* parasite causes the disease malaria.

The photograph shows a female *Anopheles* mosquito.



4.4 mm

(Source: © Dr Tony Brain / Science Photo Library)

- (a) Calculate the magnification of the *Anopheles* mosquito in this photograph.

Give your answer to one decimal place.

(1)

Answer

- (b) In 2017, an estimated 219 million people in 87 countries had malaria, and 435 000 of these people died.

Which of the following shows the percentage of infected people who died?

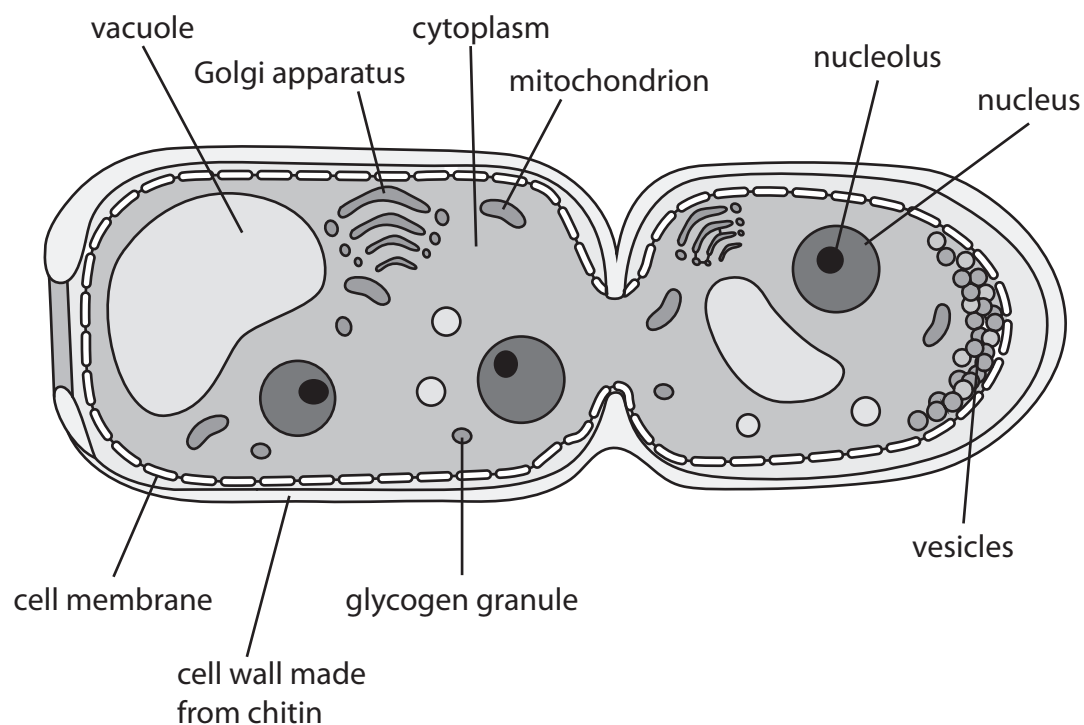
(1)

- ☐ A 0.02%
- ☐ B 0.20%
- ☐ C 99.80%
- ☐ D 99.98%



(c) The fungus *Metarhizium pingshaense* infects *Anopheles* mosquitoes.

A student drew a labelled diagram of a fungal cell that was dividing.



(i) State the function of a nucleolus.

(1)

.....

.....

.....



(ii) Give **three** differences in ultrastructure between this cell and a cell taken from the root of a plant.

(3)

1

.....

.....

2

.....

.....

3

.....

.....

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



*(d) Insecticides are chemicals used to kill insects.

Anopheles mosquitoes have developed resistance to the insecticides currently used.

Metarhizium pingshaense is a fungus that infects *Anopheles* mosquitoes.

Scientists in West Africa have genetically engineered (GE) this fungus using a gene from a spider.

When the GE fungus infects an *Anopheles* mosquito this gene becomes activated. A toxin is released, which kills the mosquito.

In an investigation, four huts each containing a group of 500 breeding pairs of mosquitoes were used.

Three different treatments used to kill mosquitoes were compared with a control group.

The walls and ceiling of the huts were sprayed with the treatment.



(Source: © Ivan Vdovin / Alamy Stock Photo)

The table shows the number of adult mosquitoes in the first two generations of offspring from each hut after spraying.

Treatment	Number of adult mosquitoes in each generation	
	First	Second
Control (no treatment)	921	1396
Insecticide	919	1353
Normal fungus	436	455
GE fungus	399	13



Evaluate these three treatments.

Use the information in the question, and your own knowledge, to support your answer.

(6)

(Total for Question 7 = 12 marks)



P 6 6 1 5 4 A 0 2 7 3 2